

Tailwind CSS answer key

Alternative class combinations are acceptable when they produce the required result using the Tailwind concepts covered in the module.

Section A: Setup and utility classes

Exercise 1: Find why Tailwind is not working

The page does not load Tailwind. Add the Play CDN script inside `<head>`:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0"
  >

  <title>Community notice</title>

  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-stone-100 p-6">
  <main class="rounded-lg bg-white p-6">
    <h1 class="text-3xl font-bold">
      Community notice
    </h1>

    <p class="mt-3 text-stone-600">
      The reading room closes at 6 PM.
    </p>
  </main>
</body>
</html>
```

The browser downloads Tailwind through the CDN script. An internet connection is required because the Tailwind code is not stored locally.

Exercise 2: Read a Tailwind class list

```
<aside
  class="
    mt-6 rounded-lg border
    border-amber-300 bg-amber-50 p-4
  "
>
  Registration closes on Friday.
</aside>
```

Utility	Effect
mt-6	Adds margin above the notice
rounded-lg	Rounds the corners
border	Adds the border width
border-amber-300	Changes the border colour
bg-amber-50	Adds a pale amber background
p-4	Adds padding inside the notice

Answers:

1. p-4 adds space inside the notice.
2. mt-6 adds space outside the notice.
3. border creates the border width.
4. border-amber-300 changes the border colour.
5. border-amber-300 only selects a colour. A border width utility such as border is still required.

Exercise 3: Put each utility on the correct element

```
<section
  class="
    rounded-lg border
    bg-white p-6
  "
>
  <h2
    class="
      text-3xl font-bold
      text-stone-900
    "
  >
    Evening photography walk
```

```

</h2>

<p>
  Meet at the east entrance at 5:30 PM.
</p>
</section>

```

`p-6` belongs on the section because the required space is between the section boundary and all its content.

Placing `p-6` on the heading would add space only around the heading. It would not provide consistent spacing around the paragraph or the complete card.

Section B: Text, spacing, backgrounds, and borders

Exercise 4: Translate CSS decisions into Tailwind utilities

```

<aside
  class="
    mb-6 rounded-lg
    border-l-4 border-amber-500
    bg-amber-100 p-4
  "
>
  <h2>Admission</h2>

  <p>
    Entry is free. Food and products
    are sold separately.
  </p>
</aside>

```

Utility mapping:

CSS declaration	Tailwind utility
<code>margin-bottom: 1.5rem</code>	<code>mb-6</code>
<code>padding: 1rem</code>	<code>p-4</code>
Amber background	<code>bg-amber-100</code>
<code>border-left: 4px</code>	<code>border-l-4</code>
Amber border colour	<code>border-amber-500</code>
<code>border-radius: 0.5rem</code>	<code>rounded-lg</code>

The Tailwind values are selected from Tailwind's spacing and colour scales. They do not reproduce the original hexadecimal values by copying them directly into the class name.

Exercise 5: Style a workshop card

```
<article
  class="
    mb-6 rounded-lg border
    border-stone-300 bg-white p-6
  "
>
  <p
    class="
      text-sm font-bold
      text-indigo-700
    "
  >
    Beginner workshop
  </p>

  <h2 class="text-2xl font-bold">
    Build your first web page
  </h2>

  <p class="mt-3 text-stone-600">
    Practise HTML structure and basic styling.
  </p>

  <p class="mt-3 font-semibold">
    Saturday, 10 AM
  </p>
</article>
```

The card-level utilities belong on the `<article>` because they control the complete component.

The text utilities belong on the individual elements whose appearance they change.

Exercise 6: Repair a broken notice

The incorrect class is:

bg-indigo-50

It should be:

bg-indigo-50

Corrected HTML:

```
<aside
  class="
    rounded-lg border-l-4
    border-indigo-500
    bg-indigo-50 p-4
  "
>
  Bring a notebook for the activity.
</aside>
```

The other styles continue to work because Tailwind processes each recognised utility independently. One invalid class does not prevent the valid classes from applying.

A useful debugging approach is to:

1. Identify which visual property is missing.
2. Locate the class responsible for that property.
3. Check its spelling and value.
4. Correct only that class.
5. Reload the page.

Section C: Flexbox and Grid

Exercise 7: Arrange a title and status on one row

```
<div
  class="
    flex items-center
    justify-between
  "
>
  <h2>Visual Studio Code</h2>
  <span>Installed</span>
</div>
```

Answers:

1. The parent `<div>` receives the Flexbox classes.
2. The `<h2>` and `<span>` become Flexbox items.
3. Adding `flex` to the `<h2>` would control the heading's own children. It would not arrange the heading beside its sibling `<span>`.

Exercise 8: Change the direction of a navigation group

Column layout:

```
<nav class="flex flex-col gap-3">
  <a href="#">Overview</a>
  <a href="#">Schedule</a>
  <a href="#">Contact</a>
</nav>
```

Row layout:

```
<nav class="flex gap-3">
  <a href="#">Overview</a>
  <a href="#">Schedule</a>
  <a href="#">Contact</a>
</nav>
```

Answers:

1. `flex` enables Flexbox on the navigation element.
2. `flex-col` changes the main direction from a row to a column.
3. `flex-col` only changes the direction of an existing Flexbox layout. It does not set `display: flex` by itself.

A row is the default Flexbox direction, so removing `flex-col` returns the links to a row.

Exercise 9: Build a card Grid

```
<section class="grid grid-cols-2 gap-4">
  <article>HTML foundations</article>
  <article>Responsive CSS</article>
  <article>Tailwind utilities</article>
  <article>JavaScript basics</article>
</section>
```

Answers:

1. The first two cards appear in the first row. The next two appear in the second row.
 2. A fifth card would begin a third row and occupy the first column.
 3. The `<section>` controls the Grid layout.
 4. Adding `grid-cols-2` to each `<article>` would not arrange the articles. The layout utility belongs on their common parent.
-

Exercise 10: Use Grid for cards and Flexbox inside each card

```
<section class="grid grid-cols-2 gap-4">
  <article
    class="
      rounded-lg border p-4
    "
  >
    <div
      class="
        flex items-center
        justify-between
      "
    >
      <h2>Quiet room</h2>
      <span>Available</span>
    </div>

    <p>Seats: 8</p>
  </article>

  <article
    class="
      rounded-lg border p-4
    "
  >
    <div
      class="
        flex items-center
        justify-between
      "
    >
      <h2>Discussion room</h2>
      <span>Occupied</span>
    </div>

    <p>Seats: 12</p>
  </article>
</section>
```

Answers:

1. The `<section>` is the Grid parent.
2. The two `<article>` elements are the Grid items.
3. The `<div>` inside each card is a Flexbox parent.
4. The `<h2>` and `<span>` inside each `<div>` are Flexbox items.

Grid controls the collection of cards. Flexbox controls the smaller title-and-status arrangement inside each card.

Section D: Responsive layouts

Exercise 11: Predict responsive behaviour

```
<section
  class="
    grid grid-cols-1 gap-4
    md:grid-cols-2
  "
>
  ...
</section>
```

Answers:

1. One column is used before the `md` breakpoint.
 2. Two columns are used at `md` and wider.
 3. `gap-4` continues to apply because it has no breakpoint prefix.
 4. `grid-cols-1` defines the base mobile-first layout.
 5. The unprefixed class already applies at the smallest size and continues to apply until another class overrides it.
-

Exercise 12: Correct a reversed responsive layout

```
<section
  class="
    grid grid-cols-1 gap-6
    md:grid-cols-2
  "
>
  ...
</section>
```

Answers:

1. `grid-cols-1` applies before a breakpoint condition is met.
 2. `md:grid-cols-2` changes the column count at `md` and wider.
 3. The narrow-screen layout is written first as the base. The wider layout is added only when the `md` condition becomes active.
-

Exercise 13: Create a responsive header

```
<header
  class="
    flex flex-col gap-4
    md:flex-row
    md:items-center
    md:justify-between
  "
>
  <div>
    <h1>Community learning centre</h1>
    <p>Weekend workshops and study groups</p>
  </div>

  <a href="#">View schedule</a>
</header>
```

The unprefixed classes apply on narrow screens:

```
flex
flex-col
gap-4
```

At `md` and wider:

- `md:flex-row` changes the direction.
- `md:items-center` aligns the items vertically.
- `md:justify-between` moves the two sides apart.

Final exercise: Build a responsive learning-space directory

Completed `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0"
  >
```

```

<title>Learning-space directory</title>

<script src="https://cdn.tailwindcss.com"></script>
</head>
<body
  class="
    bg-stone-100
    p-6 text-stone-900
  "
>
  <main class="mx-auto max-w-5xl">
    <header
      class="
        mb-6 flex flex-col gap-4
        md:flex-row
        md:items-center
        md:justify-between
      "
    >
      <div>
        <p
          class="
            text-sm font-bold
            text-indigo-700
          "
        >
          Community centre
        </p>

        <h1 class="text-3xl font-bold">
          Learning-space directory
        </h1>

        <p class="mt-3 text-stone-600">
          Find a suitable room for individual
          study or group work.
        </p>
      </div>

      <a
        href="#"
        class="
          rounded-lg bg-indigo-700
          px-4 py-2 font-semibold
          text-white
        "
      >
        View centre rules
      </a>
    </header>
  </main>
</body>

```

```

<aside
  class="
    mb-6 rounded-lg
    border-1-4 border-amber-500
    bg-amber-100 p-4
  "
>
  Rooms marked Available can be used
  without an advance booking.
</aside>

<section
  class="
    grid grid-cols-1 gap-6
    md:grid-cols-2
  "
>
  <article
    class="
      rounded-lg border
      border-stone-300
      bg-white p-6
    "
  >
    <div
      class="
        flex items-center
        justify-between
      "
    >
      <h2 class="text-xl font-bold">
        Quiet room
      </h2>

      <span
        class="
          text-sm font-bold
          text-emerald-700
        "
      >
        Available
      </span>
    </div>

    <p class="mt-3 text-stone-600">
      A low-noise room for reading
      and individual study.
    </p>

    <p class="mt-3 font-semibold">
      Seats: 8
    </p>

```

```

    </p>
</article>

<article
  class="
    rounded-lg border
    border-stone-300
    bg-white p-6
  "
>
  <div
    class="
      flex items-center
      justify-between
    "
  >
    <h2 class="text-xl font-bold">
      Discussion room
    </h2>

    <span
      class="
        text-sm font-bold
        text-amber-700
      "
    >
      Occupied
    </span>
  </div>

  <p class="mt-3 text-stone-600">
    A shared room for group activities
    and project discussions.
  </p>

  <p class="mt-3 font-semibold">
    Seats: 12
  </p>
</article>

<article
  class="
    rounded-lg border
    border-stone-300
    bg-white p-6
  "
>
  <div
    class="
      flex items-center
      justify-between

```

```

    "
  >
    <h2 class="text-xl font-bold">
      Computer room
    </h2>

    <span
      class="
        text-sm font-bold
        text-emerald-700
      "
    >
      Available
    </span>
  </div>

  <p class="mt-3 text-stone-600">
    Desktop computers with internet
    access for guided practice.
  </p>

  <p class="mt-3 font-semibold">
    Seats: 10
  </p>
</article>
</section>
</main>
</body>
</html>

```

Expected responsive behaviour

Narrow screens

- The header contents are stacked.
- The schedule link appears below the title group.
- The room directory uses one column.
- The page retains padding around its edges.

md and wider

- The header changes to a row.
- The title group and link move to opposite ends.
- The room directory changes to two columns.
- The third card continues onto the next Grid row.

Layout responsibilities

- The `<header>` controls the arrangement of the title group and link.
- The room `<section>` controls the card Grid.

- Each card's inner `<div>` controls the title-and-status row.
- The `<article>` elements contain the card appearance utilities.
- The page background and outer padding belong on `<body>`.
- The width limit and horizontal centring belong on `<main>`.